Electromagnetic Waves

The following subjects will be covered

- Power Density in EM Waves.
- Wave Polarization.
- Multi-dimensional Propagation.

EM Wave Power Density

Starting with Maxwell's equations, a formulation of the radiated power can be obtained.

$$\vec{\nabla} \times \vec{E} = -\mu \frac{\partial \vec{H}}{\partial t}$$

$$\vec{\nabla} \times \vec{H} = \sigma \vec{E} + \varepsilon \frac{\partial \vec{E}}{\partial t}$$

$$\vec{E} \cdot (\vec{\nabla} \times \vec{H}) = \sigma \vec{E} \cdot \vec{E} + \vec{E} \cdot \varepsilon \frac{\partial \vec{E}}{\partial t}$$

$$\vec{\nabla} \cdot (\vec{A} \times \vec{B}) = \vec{B} \cdot (\vec{\nabla} \times \vec{A}) - \vec{A} \cdot (\vec{\nabla} \times \vec{B})$$

$\vec{H}$ $\vec{E}$ $\vec{E}$ $\vec{H}$ $\vec{H}$ $\vec{E}$

$$\vec{\nabla} \cdot (\vec{H} \times \vec{E}) + \vec{H} \cdot (\vec{\nabla} \times \vec{E}) = \vec{E} \cdot (\vec{\nabla} \times \vec{H})$$

$$\vec{\nabla} \cdot (\vec{H} \times \vec{E}) + \vec{H} \cdot (\vec{\nabla} \times \vec{E}) = \sigma \vec{E} \cdot \vec{E} + \varepsilon \vec{E} \cdot \frac{\partial \vec{E}}{\partial t}$$

$$-\vec{\nabla} \cdot (\vec{E} \times \vec{H}) - \frac{1}{2} \mu \frac{\partial |\vec{H}|^2}{\partial t} = \sigma |\vec{E}|^2 + \frac{1}{2} \varepsilon \frac{\partial |\vec{E}|^2}{\partial t}$$

$$\frac{\partial |\vec{E}|^2}{\partial t} = \vec{E} \cdot \frac{\partial \vec{E}}{\partial t} + \vec{E} \cdot \frac{\partial \vec{E}}{\partial t} = 2\vec{E} \cdot \frac{\partial \vec{E}}{\partial t}$$

$$\vec{H} \cdot (\vec{\nabla} \times \vec{E}) = \vec{H} \cdot \left(-\mu \frac{\partial \vec{H}}{\partial t} \right) = -\mu \vec{H} \cdot \frac{\partial \vec{H}}{\partial t}$$

$$\vec{\nabla} \cdot (\vec{E} \times \vec{H}) + \sigma |\vec{E}|^2 + \frac{1}{2} \left[\mu \frac{\partial |\vec{H}|^2}{\partial t} + \varepsilon \frac{\partial |\vec{E}|^2}{\partial t} \right] = 0$$

$\frac{\frac{V}{m}\frac{A}{m}}{m}$ $\frac{W}{m^3}$ $\frac{W}{m^3}$

$\frac{W}{m^3}$ Power volume density

EM Wave Power Density

Integrating over some volume (V), and using the divergence theorem,

$$\oint (\vec{E} \times \vec{H}) \cdot \overrightarrow{ds} + \int \sigma |\vec{E}|^2 \, dv + \frac{\partial}{\partial t} \int \left(\frac{1}{2} \mu |\vec{H}|^2 + \frac{1}{2} \varepsilon |\vec{E}|^2 \right) dv = 0$$

$$\frac{V}{m} \frac{A}{m} = \frac{W}{m^2}$$

(W) Existing (Radiated) power

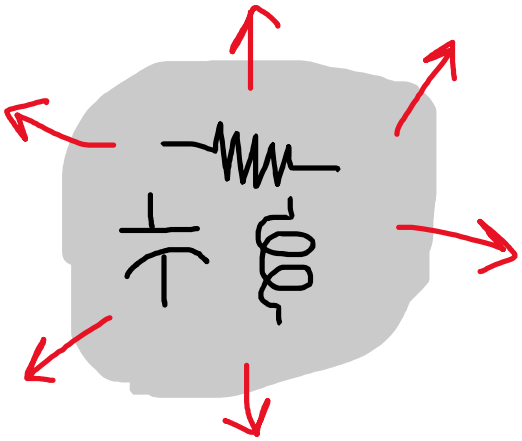
(W) Dissipated power

(W) Stored power

$$\vec{S} = \vec{E} \times \vec{H}$$

Radiated Power density
W/m^2

The Poynting vector



Power Density in a Plane Wave

Consider the following plane wave represented by E and H.

$$\vec{E} = E_o \cos(\omega t - \beta z)\, e^{-\alpha z}\ \vec{a}_x$$

$$\vec{H} = H_o \cos(\omega t - \beta z - \theta_\eta)\, e^{-\alpha z}\ \vec{a}_y$$

$$\vec{S} = \vec{E} \times \vec{H}$$

Poynting vector
(Wave power density W/m^2)

$$\vec{S} = E_o H_o \cos(\omega t - \beta z) \cos(\omega t - \beta z - \theta_\eta)\ e^{-2\alpha z}\ \vec{a}_z$$

$$\vec{S} = \frac{{E_o}^2}{|\eta|} \cos(\omega t - \beta z) \cos(\omega t - \beta z - \theta_\eta)\ e^{-2\alpha z}\ \vec{a}_z$$

Remark
This is just one specific case. For any other situation, the same procedure is used.

The time-average power density is given by

$$\vec{S}_{ave} = \frac{1}{T} \int_0^T \vec{S}\ dt\ \vec{a}_z$$

For a plane wave, $\vec{S}_{ave} = \frac{1}{2}\, Re\left[\vec{E} \times \vec{H}^*\right]$ ➡

$$\vec{S}_{ave} = \frac{{E_o}^2}{2|\eta|} \cos\theta_\eta\, e^{-2\alpha z}\ \vec{a}_z \quad (\text{W/m}^2)$$

Time-average power density

Total average power within some area $P_{ave} = \int_A \vec{S}_{ave} \cdot \overrightarrow{dA}$ (W)

Examples

(1) Consider an EM wave given by $\vec{E} = 4\ \sin(2\pi\times10^7 t - 0.8x)\ \vec{a}_z$ (V/m) travelling in a non-magnetic medium. Find ε_r, η and $\vec{S}_{ave}$.

Lossless medium $\alpha = 0$

Wave number $\beta = 0.8$ (rad/m)

Frequency $\omega = 2\pi\times10^7$ (rad/s)

Using $\beta = \dfrac{\omega}{v}$ and $v = \dfrac{c}{\sqrt{\mu_r \varepsilon_r}}$

$\Rightarrow \varepsilon_r = (\beta c/\omega)^2 \Rightarrow \varepsilon_r = 14.4$

$$\eta = \eta_o \sqrt{\frac{\mu_r}{\varepsilon_r}} = \frac{377}{\sqrt{14.4}} = 98.6\ \Omega$$

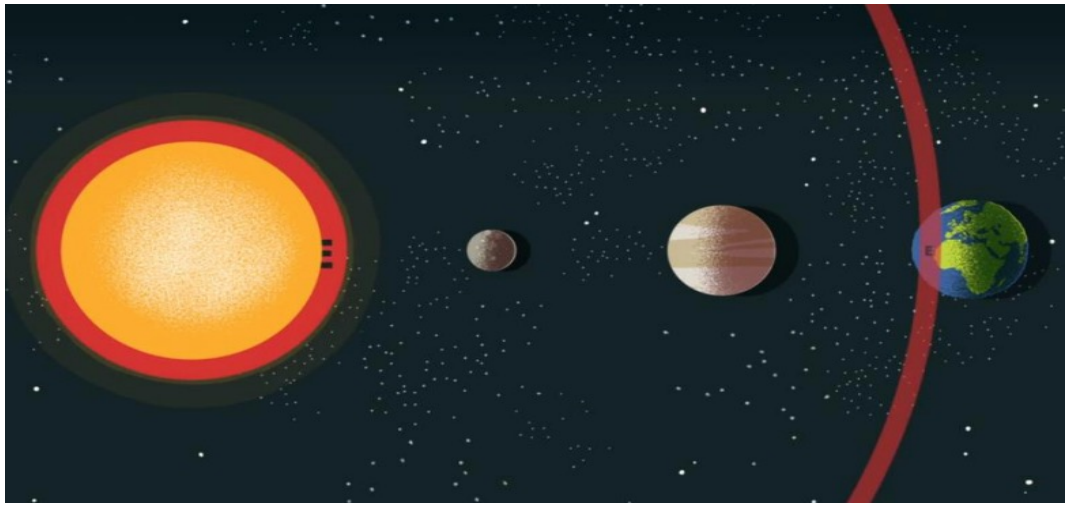
$$\vec{S}_{ave} = \frac{{E_o}^2}{2|\eta|} \cos\theta_\eta\ e^{-2\alpha x}\ \vec{a}_x$$

Impedance is real, so $\theta_\eta = 0$

$$\vec{S}_{ave} = \frac{4^2}{2(98.6)}\ \vec{a}_x = 81.1\ \vec{a}_x\ (\text{mW/m}^2)$$

Examples

(2) Find the magnitude of the electric field intensity associated with solar radiation at Earth surface. The total power radiated from the sun in about $3.8\times10^{26}\,W$ and the average distance to Earth is 150×10^{6} Km.



The solar power density at Earth surface is given by

$$S_{ave} = \frac{3\times10^{26}}{4\pi r^2} = \frac{3\times10^{26}}{4\pi(150\times10^{9})^2} = 1.06\times10^{3} \quad \text{W/m}^2$$

At Pluto surface this value is about 25.7 V/m.

The corresponding electric field intensity is

$$E_o = \sqrt{2|\eta|S_{ave}} = \sqrt{2(377)(1.06\times10^{3})} = 894 \quad \text{V/m}$$

Wave Polarization

Wave polarization is defined as the trace that the electric field vector orientation makes as the wave propagates.

$$\vec{E} = (E_o \vec{a}_E)\ e^{j(\omega t - \beta z)}$$

$$\vec{E} = (Ae^{j\phi_A}\ \vec{a}_x + Be^{j\phi_B}\ \vec{a}_y)\ e^{j(\omega t - \beta z)}$$

At $z = 0$

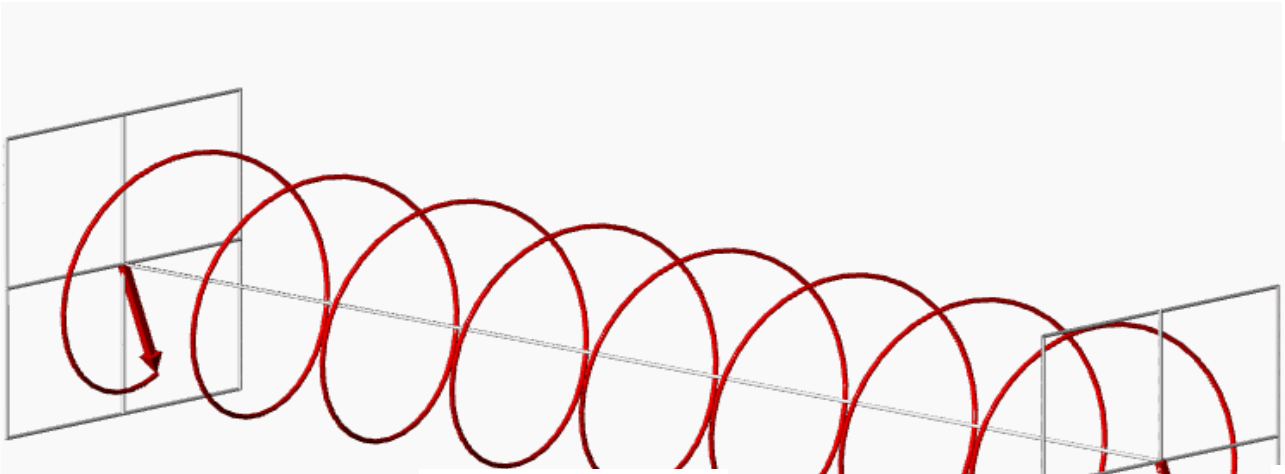
$$E_x = A\cos(\omega t + \phi_A) \qquad E_y = B\cos(\omega t + \phi_B)$$

Linear Polarization $\quad \phi_A - \phi_B = 0, \pi \quad$ or $\quad A = 0 \quad$ or $\quad B = 0$

Circular Polarization $\quad \phi_A - \phi_B = \pm\frac{\pi}{2} \quad$ and $\quad A = B$

Elliptical Polarization $\quad \phi_A - \phi_B \neq \pm\frac{\pi}{2} \quad$ or $\quad A \neq B$

The Uniform Plane-wave Solution



Linear
Polarization
Horizontal

Circular
Polarization
Right-handed

Wave Polarization: A Simple Procedure

Start by writing the electric field vector in the following format.

$$\vec{E} = (A\,\vec{a}_{\parallel} + B\,\vec{a}_{\perp})\; e^{-j\beta r}$$

Horizontal component

Vertical component

Relative to a wave going out of the page. Use right-hand rule.

Evaluate $\frac{B}{A}$

$\pm k$ — **Linear Polarization**
- Only one component
- Two components in phase

$1\angle +\frac{\pi}{2}$ LH (CCW)

$1\angle \frac{-\pi}{2}$ RH (CW)

Circular Polarization

$k\angle +\phi$ LH (CCW)

$k\angle -\phi$ RH (CW)

Elliptical Polarization

Examples

1) Discuss the polarization of the following wave.

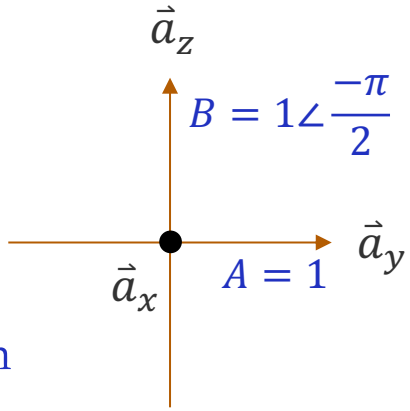
$$\vec{E} = \cos(\omega t - \beta x)\ \vec{a}_y + \sin(\omega t - \beta x)\ \vec{a}_z$$

$$\vec{E} = (A\,\vec{a}_{\parallel} + B\,\vec{a}_{\perp})\ e^{-jkr}$$

Horizontal component — Vertical component

$$\vec{a}_k = \vec{a}_x$$

$$\frac{B}{A} = 1\angle\frac{-\pi}{2}$$



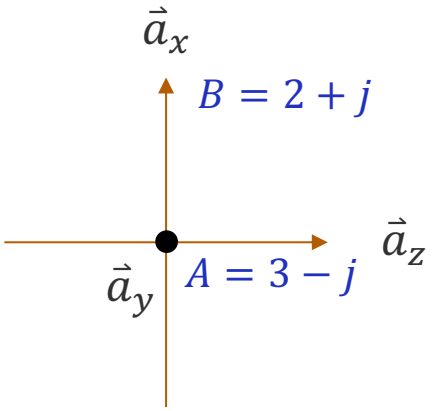
RH Circular Polarization

2) Discuss the polarization of the following wave.

$$\vec{E} = [(2+j)\,\vec{a}_x + (3-j)\,\vec{a}_z]\ e^{-j\beta y}$$

$$\vec{a}_k = \vec{a}_y$$

$$\frac{B}{A} = \frac{2+j}{3-j} = \frac{1}{\sqrt{2}}\angle\frac{\pi}{4}$$



LH Elliptical Polarization

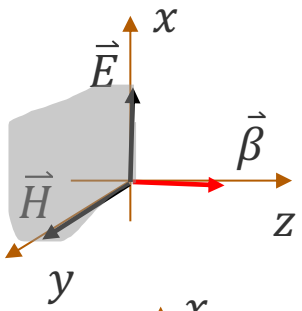
Oblique Propagation

It is possible that a plane wave propagates along a direction different from a principal axis (a combination of axis). Consider propagation in a lossless medium.

$$\vec{E} = E_o \ \vec{a}_x \ e^{-j\beta z}$$

$$\vec{H} = H_o \ \vec{a}_y \ e^{-j\beta z}$$

$$\vec{\beta} = \beta \ \vec{a}_z$$

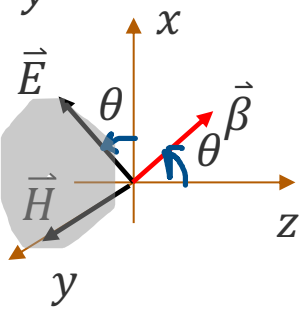


Transverse Electric Magnetic (TEM)

$$\vec{E} = E_o(\cos\theta \ \vec{a}_x - \sin\theta \ \vec{a}_z) \ e^{-j(\beta_x x + \beta_z z)}$$

$$\vec{H} = H_o \ \vec{a}_y \ e^{-j(\beta_x x + \beta_z z)}$$

$$\vec{\beta} = \beta_x \ \vec{a}_x + \beta_z \ \vec{a}_z$$

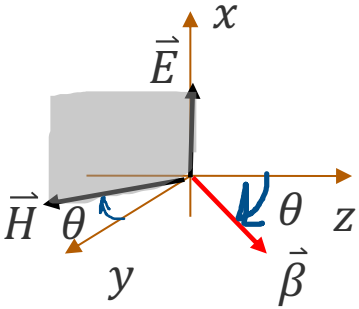


Transverse Magnetic (TM)

$$\vec{E} = E_o \ \vec{a}_x \ e^{-j(\beta_y y + \beta_z z)}$$

$$\vec{H} = H_o(\cos\theta \ \vec{a}_y - \sin\theta \ \vec{a}_z) \ e^{-j(\beta_y y + \beta_z z)}$$

$$\vec{\beta} = \beta_y \ \vec{a}_y + \beta_z \ \vec{a}_z$$



Transverse Electric (TE)

Examples

1) Discuss the wave direction. $\vec{E} = E_o \vec{a}_y \, e^{j\omega t} \, e^{-j(x+2z)/\sqrt{2}}$ (V/m)

Transverse Electric (TE)

$\beta_x = 1/\sqrt{2}$ $\beta_z = 2/\sqrt{2}$ $\vec{\beta} = \frac{1}{\sqrt{2}} \vec{a}_x + \frac{2}{\sqrt{2}} \vec{a}_z$ rad/m

2) The complex electric field intensity of a time-harmonic wave travelling in a source-free free space is given by (distance in meters)

$$\vec{E} = (80 \, \vec{a}_x - 60 \, \vec{a}_z) \, e^{-j(6x+8z)} \text{ (V/m)}$$

What is the wavelength in the wave propagation direction?

$\beta_x = 6$ $\beta_z = 8$ $\beta = \sqrt{6^2 + 8^2} = 10$ rad/m $\lambda = \frac{2\pi}{k} = 0.628$ m

What is the angle the wave propagation direction makes with z?

$\tan\theta = 6/8$ $\theta = 36.9°$

Transverse Magnetic (TM)

Polarization?

Q & A

Send me an email if you have any questions.